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Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

National Science Foundation

Submitted to:

National Science Foundation

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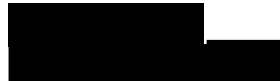
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Maintaining and Accelerating the U.S. Lead in AI

Introduction

The United States leads the world in artificial intelligence (AI) innovation, driven by a dynamic private sector, powerful research institutions, and strategic governmental support. As AI technologies continue to evolve, it is crucial to maintain and accelerate U.S. leadership by promoting accelerated AI adoption and fostering an environment conducive to innovation and global competitiveness. Our response outlines strategic policy recommendations intended to sustain and enhance the U.S. lead in AI.

Our recommendations can be summarized as “prioritize guardrails and remove barriers.” To accelerate AI innovation and ensure broad adoption, the United States should embrace a decentralized governance model that empowers industries to develop sector-specific standards, with the government providing broad policy guidance to promote transparency, security, and risk management. A flexible, sector-driven approach to AI governance would allow industries to create tailored frameworks while ensuring alignment with national priorities.

This effort would include:

- **Encouraging industry-led standard setting.** Sectors should develop AI governance frameworks suited to their operational and ethical considerations, ensuring practical and effective compliance.
- **Establishing broad federal AI policy guidelines.** The government can provide overarching principles—such as risk transparency, security, and responsible AI use—while allowing industries to determine implementation specifics.
- **Enhancing interagency coordination.** AI policy councils should align governance approaches across sectors, ensuring consistency while allowing for sector-specific flexibility.
- **Supporting public-private partnerships.** Collaboration between government, industry, and academia can ensure that voluntary standards align with national interests, including economic growth, workforce readiness, and national security.

The following sections provide more specific recommendations.

Procurement

To maintain and accelerate U.S. leadership, it is vital to employ procurement approaches that allow government to keep pace with rapid advancements in the field. New developments occur every month; new milestones are achieved each quarter. To be innovative with AI technologies, the government must be innovative with its approach to procurement.

Accelerate AI Innovation by Innovating With Procurement

To take advantage of AI, the government should provide fast-track acquisition pathways for AI tools and capabilities.

The General Services Administration (GSA) Schedule and the Department of Defense (DoD) Other Transaction Authority (OTA) are effective examples of how policy can shorten procurement lead times. Those

mechanisms should be expanded—or similar mechanisms should be created—so that agencies across government can rapidly adopt and take advantage of AI solutions.

The government should also provide incentives and partnerships that encourage the development of innovative approaches. The Small Business Innovation Research (SBIR) program is an excellent example of how innovation can be driven through policy. The Department of Homeland Security (DHS) Science and Technology Directorate's Silicon Valley Innovation Program (SVIP) also encourages innovation by offering a streamlined process to prototype cutting-edge solutions. These are effective models that should be expanded upon, allowing more private firms to accelerate AI adoption and use.

In addition, the government should adopt policies that encourage the following improvements to procurement processes:

- Greater use of pilot-first procurement models that allow agencies to test, refine, and validate AI solutions before committing to full-scale implementation
- Standardization of AI risk evaluation criteria across agencies to create consistent expectations for vendors and accelerate procurement of proven models and solutions
- A focus on outcome-based performance metrics rather than rigid compliance checklists so that vendors are motivated to continuously improve and are aligned with desired agency outcomes
- Eliminating workforce minimum required staffing levels so that vendors can employ AI solutions creatively and harness their full benefit
- Improving compliance and oversight reporting by shifting to automated AI-based validation instead of manual reviews

- Streamlining the Federal Acquisition Regulation (FAR) to allow agencies to buy at the speed of need

Incentivize Investment in AI Innovation

AI systems create enormous efficiency gains. U.S. private industry is reaping these benefits and replacing time-consuming, manual processes with automated routines. However, similar gains come more slowly to the federal government because existing policy *inhibits* on-contract innovation. Most contracts pay based on hours worked. As a result, contractors are not incentivized to introduce AI solutions that increase efficiency because reducing the hours worked will reduce the size of their contract.

Policies should be introduced that allow more flexibility on the part of contractors. If they were paid based on outcomes achieved or objectives obtained—perhaps using an objectives and key results (OKR) structure or milestone-driven payments—they would be *incentivized* to introduce innovations that result in efficiencies and produce improved outcomes for the American people. Additionally, establishing clear guidelines for converting existing AI-enabled contracts to outcome-based models will help agencies adopt this model effectively while minimizing risks.

Clarify AI Use in Proposal Processes

No clear standard exists for when and how it is acceptable to employ AI in drafting proposals for federal work. This leads to waste and inefficiency, especially as AI becomes increasingly integrated into standard business productivity tools, such as the Microsoft Office suite. The government will benefit from establishing clear policies for how contractors can or cannot use AI in their proposal efforts. Such policies should differentiate between AI-

generated content and AI-assisted efficiency tools so that AI use for non-substantive enhancements like formatting, grammar checks, and data organization are always acceptable. Clear, realistic guidelines on permissible AI use across the government will accelerate and enhance procurement processes.

Eliminate Restrictions that Block Technical Collaboration

The rapid advancement of AI technology combined with existing procurement policies makes it more difficult for government agencies to acquire the best solution. Restrictions that prevent early engagement between potential vendors and government stakeholders inhibit evaluation of AI solutions and exploration of alternatives.

Policies should be introduced that allow for structured pre-solicitation phases and facilitate technical collaboration between potential vendors and agency stakeholders. This will reduce the risk of undefined requirements—which often lead to wasteful contract modifications—and ensure that proposals are more accurate and complete, permitting agencies to make more informed procurement decisions.

Prevent Model Lock-In

Policies should be introduced that avoid AI vendor and model lock-in by encouraging flexibility in the use of specific AI models. The foundation model ecosystem is changing rapidly. New, powerful proprietary and open-source models are becoming available every quarter. To allow agencies to take advantage of the best technologies, government policies should be introduced that incentivize agencies to preserve flexibility and that encourage the use of contractors who can support new and different models and vendors as they become available.

Education and Workforce

To maintain and accelerate U.S. leadership, AI solutions must be adopted and used. Currently, the most significant limitation on their use is education and training. The federal government can accelerate AI innovation by introducing policies that encourage education and training for AI use, such as the following:

- First, the government should provide expanded funding for AI education and training, both for citizens and government employees, to encourage the growth of an AI-ready workforce. Funding should be flexible and outcome oriented so that industries and agencies can adapt specific training and education to their needs and employ approaches such as workshops, online courses, and certifications focused on the practical uses of AI, its applications, and ethical considerations.
- Second, public-private partnerships should be created with the mission of building and maintaining a high-proficiency AI workforce. These partnerships should be aligned with specific industries so that skills development is rapid and responsive to the needs of those industries. This is essential to ensure the workforce keeps pace with AI advancements, addressing both talent and skill set shortages so industries can fully leverage AI-driven innovation and productivity gains.

These steps will help ensure that, as the field advances, the United States maintains its leadership position in AI by investing in talent. A strong domestic AI workforce supports long-term innovation and economic growth, securing America's position as a global leader.

Application and Use

AI applications face hurdles because many citizens lack familiarity with the underlying technology and are apprehensive about the associated risks. To spur adoption and ensure the United States maintains its lead in the field, policy must address these concerns.

Establish a Public AI Best Practices Repository

The federal government should create a centralized, publicly accessible AI Best Practices Repository where agencies can share case studies, procurement guidelines, implementation approaches, and risk management strategies.

This repository should include:

- Case studies detailing the approach taken and beneficial outcomes achieved
- Procurement and implementation guides
- Risk management frameworks (RMFs) and strategies

Facilitate Access to AI Resources and Data

The quality of AI outputs is largely a question of data, both the training data and the data available for analysis and computation. Currently, AI innovation in the federal government is constrained by the lack of data sharing between and among agencies and departments. To innovate faster, this must change.

The federal government can encourage broader AI application and use by promoting the sharing of data held by federal agencies, especially data that are not subject to intellectual property (IP) or privacy restrictions. This will provide other agencies, private industry, and researchers with valuable resources for developing new AI capabilities and technologies, especially in critical fields like healthcare. Modern approaches to data transparency and

integration will allow AI solutions to reflect the full potential of the federal government and industry partners. This will increase confidence, improve the quality of AI results, and encourage broader adoption.

Encourage Responsible AI

For critical sectors (e.g., healthcare, finance, transportation), errors are unacceptable, and risks must be managed. In these sectors, guidelines for responsible AI must be developed and employed. AI has great potential to improve outcomes and drive efficiency; however, to ensure that potential is realized, citizens must have confidence in the AI systems that are developed and used. Accordingly, the government should prompt the development of guidelines and regulations for responsible development and use of AI systems. A balanced approach to regulation will increase confidence, accelerate adoption, and promote the necessary innovation to keep the United States in its global leadership position.

Accelerate ATO Processes for AI Applications

To encourage AI adoption and use across government, mechanisms must be put in place to allow Authority to Operate (ATO) processes to move more quickly. Current ATO processes are costly and time consuming. They disincentivize the development of AI innovations that would otherwise benefit government and citizens.

Improvements to ATO processes should include:

- Preapproved AI security models and reference architectures
- Fast-track approval pathways for low-risk AI applications
- Tiered, risk-based ATO approvals

These approaches will accelerate the development and deployment of AI solutions and incentivize experimentation

with innovative AI tools, resulting in broader use of AI across government.

Open-Source Development

Open-source tools and techniques accelerate innovation and provide a valuable stimulant to research and development. Accordingly, the federal government should view open-source approaches strategically and take several supporting policy actions.

Promote Open-Source AI Initiatives

More so than other technologies, AI is a “public good” challenge. Effective AI technologies require high-quality data for training and testing; much of those data are publicly available, but they need to be accessible in formats that allow private companies and the federal government to use them and take advantage of them. Policies should be crafted that encourage the development and use of these formats. Google’s TensorFlow project is a valuable example of how this can be done and how open-source approaches can stimulate AI research and development.

Support Collaborative Research

Innovation occurs more rapidly when ideas and concepts jump from one discipline to another, providing new ways to look at old problems. The federal government should therefore introduce policies that facilitate partnerships between industry, academia, and government that address national challenges using open-source AI solutions.

The Defense Innovation Unit (DIU) is an excellent example of how these kinds of partnerships can be employed advantageously. DIU (1) works with commercial industry to identify compelling solutions that have potential, (2) funds pilot contracts, and (3) facilitates rapid procurement if those pilots prove fruitful. The federal government ought to introduce

policies that allow similar partnerships and rapid procurement for other agencies and programs. It should also encourage the use of open-source techniques in these procurements so that other agencies and industries can rapidly benefit.

Explainability and Assurance of AI Model Outputs

The reasoning processes of AI systems are regularly criticized as “opaque.” The lack of explainability discourages adoption and use, ultimately limiting innovation and endangering the U.S. lead in this important field. Policies can be tailored to stimulate innovation while increasing explainability. This will spur adoption and encourage further innovation, creating a virtuous cycle.

Standardize Explainability Metrics

The federal government should collaborate with private industry to develop industry-wide standards for measuring and reporting the explainability of AI models. This will move explainability from a qualitative discipline to a quantitative one. With measurable standards, the transparency and interpretability of AI systems will increase, encouraging broader adoption and use.

Some tools, such as LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations), are already being used to provide insights into model predictions. These tools are examples of how to provide greater transparency into AI outputs and illustrate a path forward. However, the lack of standardized metrics for explainability poses challenges in comparing and validating AI models across different applications and industries. Federal encouragement for industry-wide standards will address this challenge.

Invest in Explainable AI (XAI) Research

Policies should be introduced to support research efforts to develop inherently interpretable AI models and improve existing techniques for model explainability. This is a strategic investment that can accelerate adoption and use of AI solutions in the essential fields of healthcare, finance, and autonomous systems where understanding the rationale behind AI decisions is crucial.

The Defense Advanced Research Projects Agency's (DARPA's) XAI program is a leading example of work in this field. It aims to create AI systems that make decisions that can be understood and trusted by human users. Policy should encourage additional programs and avenues of research in this area so that government and private industry can collaborate on alternative approaches and pursue multiple promising pathways in parallel.

Intellectual Property

As AI adoption accelerates, it is critical to establish clear IP guidelines to address ownership, licensing, and compliance. Clear guidance is required to make existing laws more effective and help accelerate innovative use of AI technologies.

IP guidance should include the following:

- Frameworks that distinguish between AI-generated and human-created content to ensure clarity on copyright eligibility, establish consistent attribution policies, and specify disclosure standards
- Adaptation of existing contractual frameworks—such as the FAR—that stimulate private investment in AI solutions and retention of limited IP protections while safeguarding the federal government's rights

- Clarifying IP protections for data—with appropriate differentiation between public, government-owned, and proprietary datasets—so that (1) AI companies and solution providers can be confident about the scope of their IP rights and the risks of third-party data and (2) pre-trained models can be well understood and appropriately mitigated

Risks, Regulation, and Governance

Whether data privacy or other risks, when promoting AI innovation, effective risk management is crucial to ensure AI systems are developed and deployed responsibly and sustainably. However, it is essential that risk management and any associated regulations accelerate—and do not inhibit—innovation.

Develop Adaptive Regulatory Frameworks

Policy should encourage the development of regulatory frameworks that can adapt to the rapid pace of AI innovation. The National Institute of Standards and Technology (NIST) AI RMF is a good example of a standard that manages AI risk and provides a foundation for governance while stimulating—not stifling—innovation. Policy should encourage use and continued enhancement of NIST's AI RMF and similar frameworks to ensure effective risk management with AI solutions.

Encourage Automated Mechanisms for Regulatory Compliance

To ensure regulatory frameworks like the NIST AI RMF accelerate innovation, the federal government should encourage the development of automated mechanisms for validating compliance. Like automated mechanisms that validate software development processes and streamline the

ATO process in the federal government today, the introduction of automated mechanisms for AI compliance will stimulate progress and spur innovation. It will also accelerate AI adoption and use.

Establish AI Governance Bodies

The federal government should encourage the formation of governance bodies that leverage public-private partnerships to oversee the development of regulatory frameworks. The National AI Advisory Committee is an example of how this can be done; it should be adapted and configured to reinforce the current administration's priorities and to maintain the U.S. lead in AI innovation. If necessary, subcommittees could be formed to address certain areas of AI governance, regulation, and risk management.

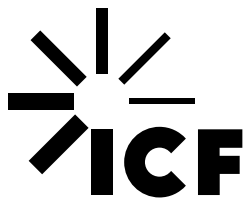
Establish AI Evaluation and Validation Standards

Developing standardized frameworks for AI performance and risk assessment can enhance innovation by providing clear guidelines without imposing restrictive regulations. This approach ensures AI applications are effective and reliable, supporting the administration's goal of maintaining global AI dominance.

Conclusion

Maintaining and accelerating the U.S. lead in AI requires a comprehensive approach that fosters innovation by creating appropriate guardrails that unlock the creativity of private industry, federal agencies, and citizens. By implementing the policy recommendations outlined in this response, the United States can create a thriving AI ecosystem that drives economic growth, enhances national security, and improves the quality of life for its citizens.

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